

Pokhara University
Faculty of Management Studies

Course Code: CMP 274
Course Title: **Numerical Methods**
Nature of the course: Theory + Practical
Year: Second
Level: Bachelor
Semester: IV

Full marks: 100
Pass marks: 45
Credit Hrs: 3
Total periods: 48 hours
Program: BCSIT

1. Course Description

This course provides a comprehensive exploration of numerical methods, equipping students with the necessary tools to solve mathematical problems encountered in computer system. The curriculum is structured around the foundation principles of Numerical Methods, integrating theoretical understanding with practical implementation. It also includes the use of methods like Bisection, Newton-Raphson methods to estimate the roots. It covers the concept of interpolation, Integration, Moreover, it covers solving ordinary and partial differential equations.

2. General Objectives

The general objectives of this course are:

- To provide students with fundamental numerical techniques, equipping them for effective problem-solving
- To equip students with the proficiency to analyze, implement, and execute numerical algorithms

3. Methods of Instruction

Lecture, Tutorial, Discussion, Lab works

4. Course Contents

Specific Objectives	Contents
- Understand the types of errors occurred in numerical problems - Explore different tools to compute numerical methods	Unit I: Introduction (2 hrs) 1.1 Errors in Numerical Calculations 1.2 Computer and Numerical Software(C,GNU Octave/Matlab) 1.3 Mathematical Preliminaries(Taylor Series)
- Understand the types of bracketing and non-bracketing methods to solve algebraic and transcendental equations - Calculate roots by using iterative approach	Unit II: Solution of non-linear equations (8 hrs) 2.1 Bisection Method 2.2 False position method 2.3 Newton-Raphson Method 2.4 Secant Method 2.5 Fixed point iteration Method



- Describe interpolation - Understand the types of interpolating techniques to find interpolating polynomial - Implement Least square method	Unit III: Interpolation & Approximation (8 hrs.) 3.1 Linear interpolation 3.2 Lagrange's interpolation 3.3 Newton's divided difference method 3.4 Newton's forward difference interpolation 3.5 Newton's Backward difference interpolation 3.6 Least square method (fitting a straight line & Non-linear curve fitting)
- Implement numerical differentiation methods - Explain Newton Cote's integration - Explain Romberg and Gauss quadrature integration	Unit IV: Numerical Differentiation & Integration (7 hrs.) 4.1 Forward, backward, and central difference methods 4.2 Trapezoidal rule 4.3 Simpson's rule 4.4 Romberg integration 4.5 Gaussian Integration (two point & three point formula)
- Understand the solution of ordinary differential equation - Implement techniques to solve higher order equations (second order) - Solve BVP by shooting method	Unit V: Solution of Ordinary Differential Equations (8 hrs.) 5.1 Euler's method 5.2 Heun's method 5.3 Runge-Kutta Method (RK-4) 5.4 Solution of Higher order equations 5.5 Boundary Value Problems (Shooting Method)
- Describe existence and behavior of solutions in numerical methods - Implement elimination, factorization, iterative methods to solve linear equations - Calculate Inverse of a matrix by Gauss-jordan elimination - Understand power method to evaluate eigen value and eigen vector	Unit VI: Solution of Linear algebraic equations (10 hrs.) 6.1 Existence of solution of linear equations 6.2 Elimination methods (Gauss-elimination, Gauss-Jordan, Partial pivoting) 6.3 Factorization methods (Doolittle, Crout, Cholesky's method) 6.4 Iterative methods (Gauss-Siedel, Gauss-Jacobi method) 6.5 Inverse of a matrix 6.6 Power method for eigen value and eigen vector
Describe second order PDE	Unit VII: Solution Of Partial Differential



- Understand Different types of PDE(hyperbolic,parabolic, elliptical) - Implement Laplace and poisson's equation	Equations (5 hrs.) 7.1 Introduction to partial differential equations 7.2 Laplace equation 7.3 Poisson's equation
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Note: The figures in the parentheses indicate the approximate periods for the respective units.

5. Laboratory Works

Use of Gnu Octave/Matlab/C/C++ high level programming language for applied numerical analysis.

- Solve non-linear equation using bisection method, secant method, Newton-Raphson method.
- Interpolation using Lagrange's interpolation and Newton's Interpolation
- Use of Newton cote's integration formula
- Numerical Differentiation using forward, backward and central difference quotient
- Program to solve euler's method, heun's method and classical method
- Solution of Partial Differential equation

6. Evaluation system and Students' Responsibilities

Evaluation System

In addition to the formal exam(s), the internal evaluation of a student may consist of quizzes, assignments, lab reports, projects, class participation, etc. The tabular presentation of the internal evaluation is as follows.

Internal Evaluation	Weight	Marks	External Evaluation	Marks
Theory		30	Semester-End examination	50
Attendance & Class Participation	10%			
Assignments	20%			
Presentations/Quizzes	10%			
Internal Assessment	60%			
Practical		20		
Attendance & Class Participation	10%			
Lab Report/Project Report	20%			
Practical Exam/Project Work	40%			
Viva	30%			
Total Internal		50		
Full Marks: 50 + 50 = 100				

Student's Responsibilities



Each student must secure at least 45% marks separately in internal assessment and practical evaluation with a minimum of 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such score will be given NOT QUALIFIED (NQ) to appear the Semester-End Examinations. Students are required to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

7. Prescribed Books and References

Text Book

1. Balagurusamy, E. (2012). Numerical Methods. McGraw-Hill Education

References Books

- Dukktipati, R. V. (2023). Numerical Methods Fundamentals. Mercury Learning & Inform.
- Gerald, C. F., & Wheatley, P. O. (2004). Applied Numerical Analysis. Pearson/Addison-Wesley.
- Cheney, E. W., & Kincaid, D. (2020). Numerical Mathematics and Computing. Brooks/Cole.
- Sastry, S. S. (2022). Introductory methods of numerical analysis. Prentice-Hall of India.

